Introduction to C Programming

Lecture 3

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What we will learn

- What is Loop

What we will learn

- What is Loop
- While Loop

What we will learn

- What is Loop
- While Loop
- For Loop

What we will learn

- What is Loop
- While Loop
- For Loop
- Do While Loop

Loop

- Repetition statements allow us to execute a statement multiple times

Loop

- Repetition statements allow us to execute a statement multiple times
- Often they are referred to as loops

Loop

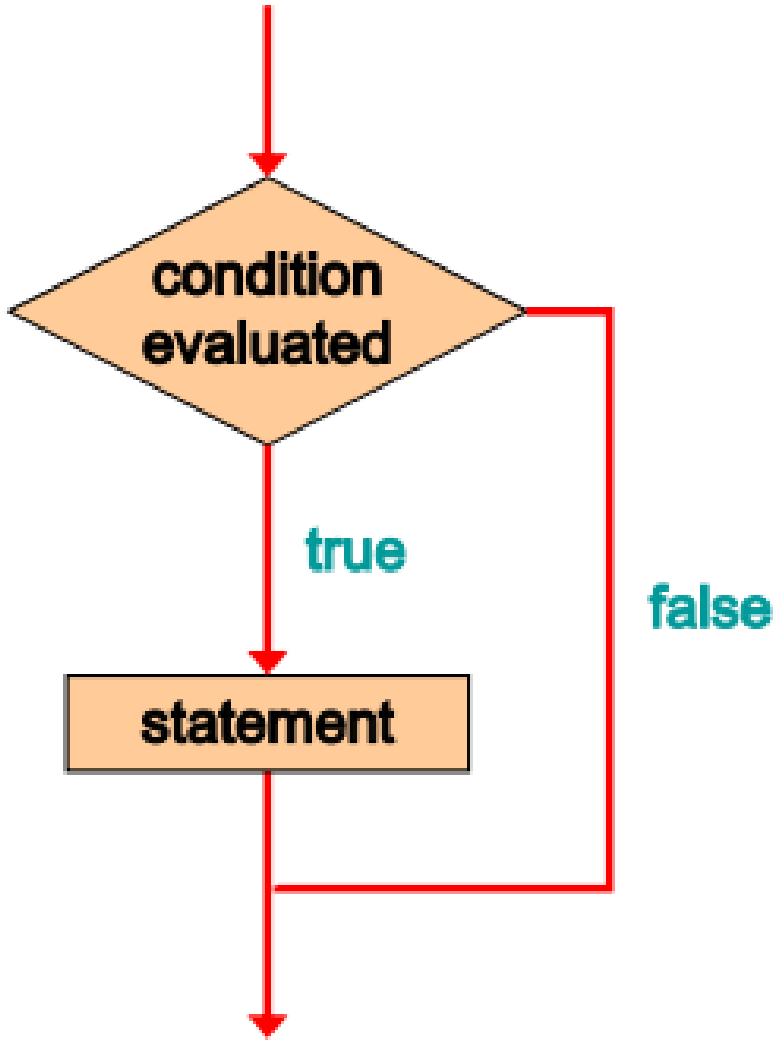
- Repetition statements allow us to execute a statement multiple times
- Often they are referred to as loops
- C has three kinds of repetition statements
  - while loop
  - for loop
  - do while loop

while loop

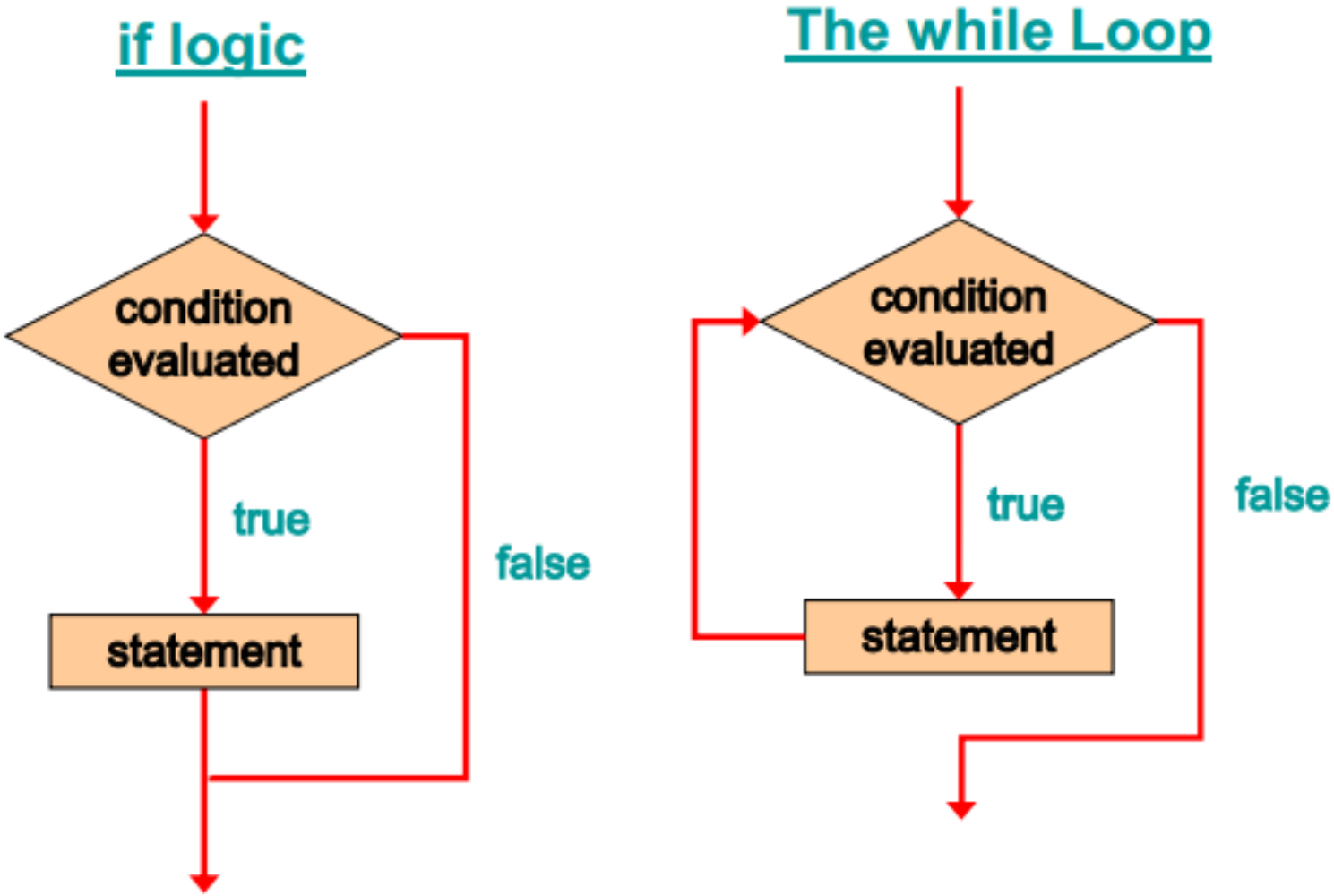
```
while ( condition )
    statement;
```

```
while ( condition ){
    statement1;
    statement2;
}
```

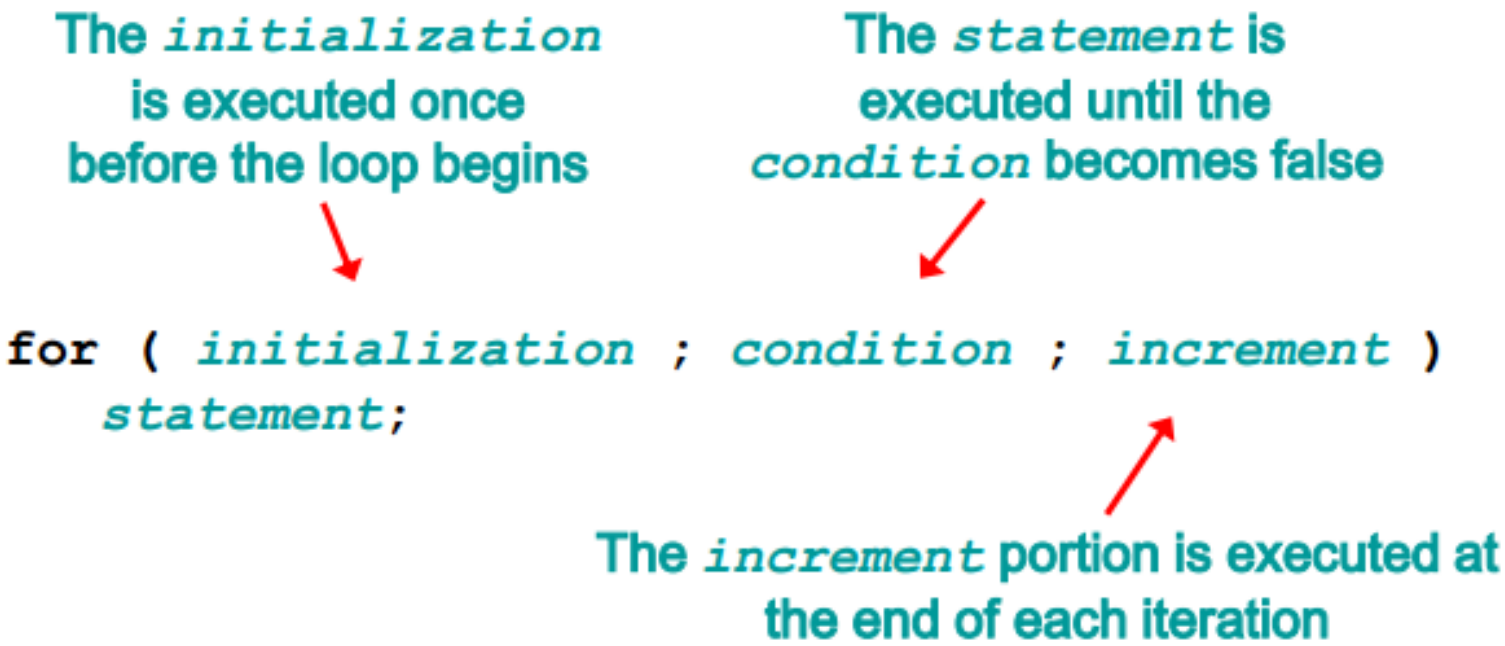
Logic of if statement



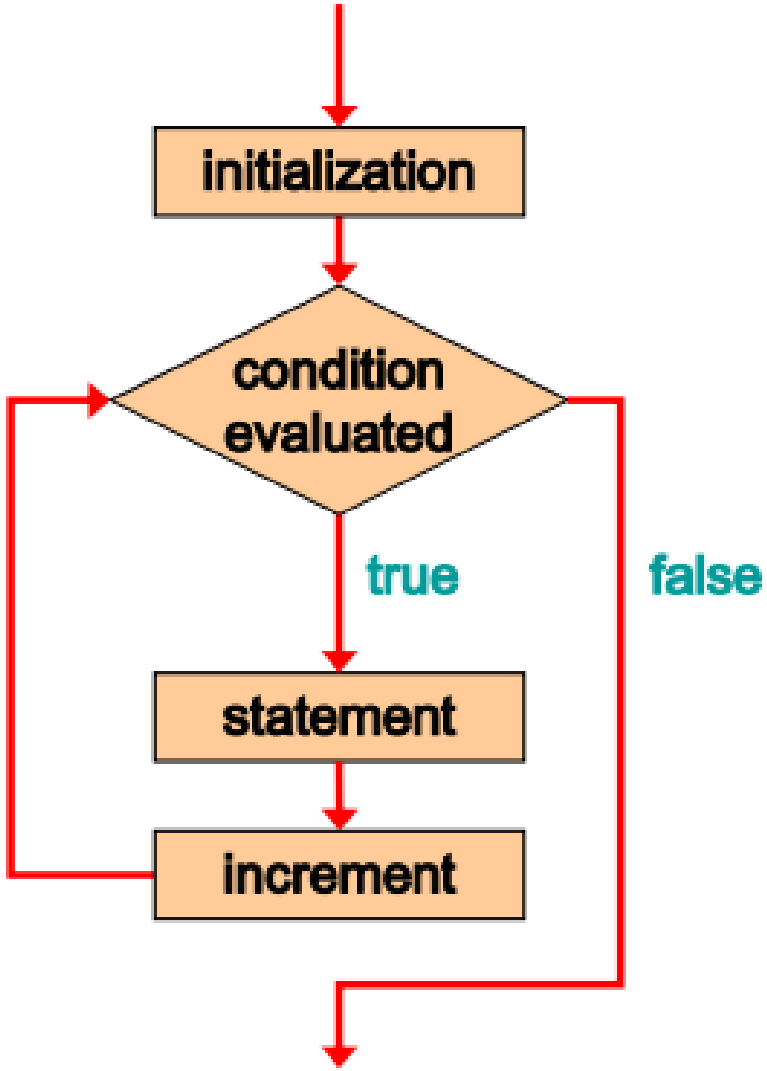
Logic of while loop



for loop


The initialization is executed once before the loop begins
The statement is executed until the condition becomes false
for (initialization ; condition ; increment)
statement;
The increment portion is executed at the end of each iteration

Logic of for loop



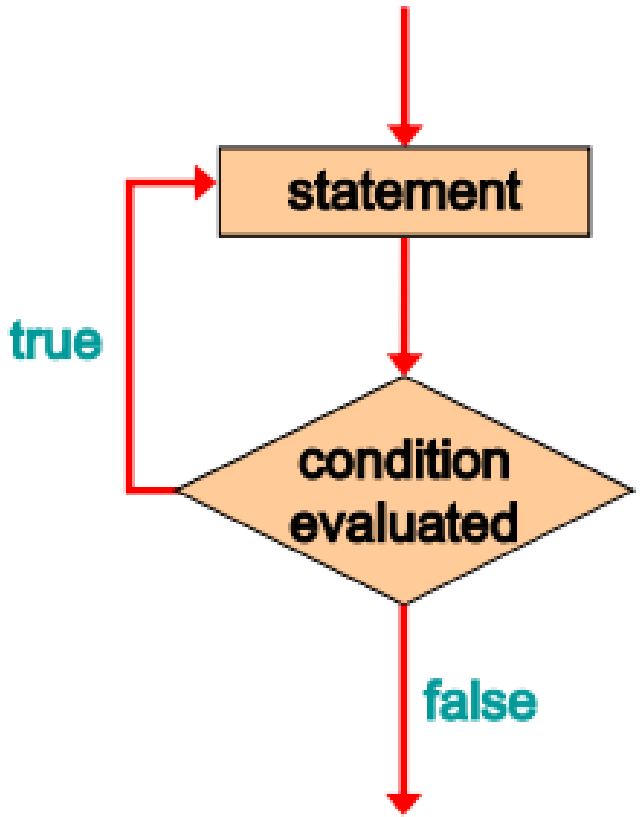
Equivalence of for loop and while loop

```
initialization;
while ( condition )
{
    statement;
    increment;
}
```

do while loop

```
do
{
    statement;
}
while ( condition );
```

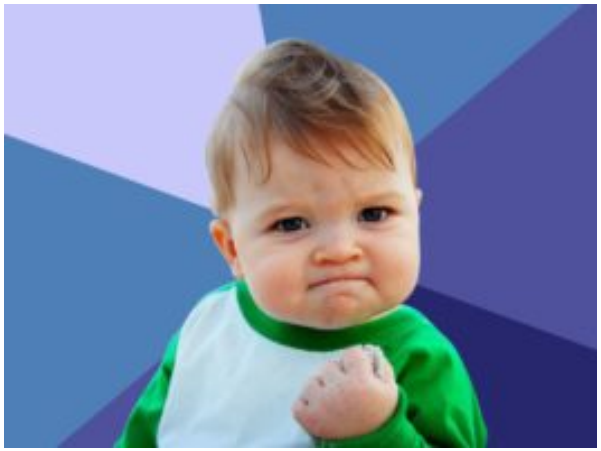
Logic of do while loop



Problems

- Print first n numbers.
- Print odd numbers up to n.
- Print even numbers up to n.
- Print summation of first n numbers.
- Print summation of all odd numbers up to n.
- Print summation of all even numbers up to n.
- Print factorial of n
- Print x^n , where x and n are integers
- Print first n Fibonacci numbers.

End



Thank You :)